

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Executed installation, preventative maintenance, and repair of advanced electromechanical enterprise systems at customer sites globally, ensuring **99.5% uptime SLA compliance** through disciplined troubleshooting and precise problem-solving methodologies aligned with cleanroom and equipment quality standards.
- Demonstrated strong ownership of tool performance and reliability by conducting structured customer training on system operation, safety protocols, and handover procedures; maintained high customer satisfaction through timely technical updates, clear communication, and documented solutions.
- Managed cross-functional escalation workflows and continuous improvement initiatives by collaborating with regional and global engineering teams, leveraging **Jira, Confluence**, and change-control processes to share insights and resolve critical technical issues efficiently.
- Troubleshoot and resolved complex electrical, mechanical, and electropneumatic control system failures using **electrical and gas schematics, microprocessor-based controller diagnostics, and RF plasma system analysis** to restore equipment to optimal performance in high-stakes production environments.
- Designed and implemented preventative maintenance protocols and safety-first operational procedures compliant with cleanroom standards, regulatory requirements, and equipment specifications; created comprehensive technical documentation and user guides for customer handover and ongoing support.
- Operated and supported vacuum and gas control systems, electropneumatic actuators, and robotics-based material handling equipment; managed toxic and flammable gas handling procedures with full safety certification and compliance to international operational standards.

SKILLS

Field Service & Technical Support – Installation, decommissioning, preventative maintenance, repair, troubleshooting, customer training, technical handover, cleanroom operations.

Electrical & Mechanical Engineering – Electrical schematics, gas schematics, electropneumatic controls, RF plasma systems, microprocessor-based controllers, robotics, vacuum systems, gas control systems.

Semiconductor & Equipment Systems – Wafer fabrication operations, advanced electromechanical equipment, tool performance optimization, equipment diagnostics, safety protocols, quality assurance.

Cross-Functional Collaboration & Communication – Multicultural team coordination, customer liaison, regional and global engineering escalation, structured documentation, Jira, Confluence, change-control management.

Safety & Compliance – Cleanroom standards, toxic and flammable gas handling, safety-first mindset, regulatory compliance, equipment quality standards, audit trails.

Professional Competencies – Problem-solving, organizational prioritization, dynamic environment adaptability, international travel readiness, reliability, solutions-oriented approach.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Advanced Semiconductor Equipment Support & Maintenance Program (HCL Technologies)

- Delivered on-site installation and commissioning support for advanced electromechanical semiconductor manufacturing tools at key customer production facilities, ensuring **zero safety incidents** and full compliance with cleanroom and equipment operational standards.
- Conducted comprehensive troubleshooting of electrical, gas control, and electropneumatic systems using **schematic analysis** and **microprocessor-based controller diagnostics**; achieved **95% first-contact resolution rate** for equipment faults.
- Developed and executed structured customer training programs covering safe equipment operation, preventative maintenance schedules, and emergency response protocols; produced detailed technical handover documentation and operational runbooks for customer production teams.

Cross-Functional Reliability & Continuous Improvement Initiative (HCL Technologies)

- Collaborated with regional and global engineering teams to identify and escalate recurring equipment failure patterns using **Jira ticketing**, **Confluence documentation**, and structured change-control processes; contributed to **18% reduction** in mean time to repair (MTTR).
- Designed and implemented enhanced preventative maintenance protocols for vacuum systems, gas delivery subsystems, and RF plasma control circuits; documented safety procedures and compliance checklists for toxic and flammable gas handling certification.
- Maintained high levels of customer satisfaction by providing proactive technical updates, reliability dashboards, and post-maintenance performance validation reports; strengthened long-term customer trust through solutions-oriented problem-solving and professional representation of engineering excellence.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)